FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 — SUB-STRAND 1.3: CELL BIOLOGY	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

This Final Explanation is your opportunity to show everything you have learned across this unit. You will answer the driving question using evidence from your investigations, observations, models, and class discussions. Read the driving question carefully, then work through each section below. Write in full sentences and use correct biological vocabulary. Where prompted, include labelled diagrams to support your explanation. A strong Final Explanation does not just list facts — it tells a clear, connected story that goes from evidence to reasoning to conclusion. Refer back to your Driving Question Board (DQB) notes, your cell model diagrams, and your microscope observations throughout this task.

Section 1: What Is a Cell? — Identifying the Shared Foundation	
Before you can explain why cells look so different, you need to establish what ALL cells have in common. Using the anchoring phenomenon as your starting point — the blood smear from the Kisumu county hospital and the sukuma wiki leaf slide — answer the following: **(a) List and describe the THREE structures that every cell, whether plant or animal, prokaryote or eukaryote, must have to be considered a living cell.** **(b) Explain what each shared structure DOES — focus on its**	Even though the red blood cell, the white blood cell, and the sukuma wiki leaf cell look strikingly different, they are all called cells because they all share three fundamental features: a cell membrane, cytoplasm, and genetic material (DNA). The cell membrane is a thin, flexible boundary made of a phospholipid bilayer that surrounds every cell. It controls what enters and leaves the cell — for example, it allows oxygen and glucose to enter a red blood cell while keeping unwanted substances out. The cytoplasm is the jelly-like fluid that fills the cell and provides the medium in which all chemical reactions of life (metabolism) take place. Genetic material (DNA) carries the instructions that tell the cell which proteins to make and how to function. Even the red blood cell, which loses its nucleus as it matures, originally contained DNA and still carries the instructions it needed to develop its final shape. These three features — membrane, cytoplasm, and DNA — are the universal hallmarks of life at the cellular level. No matter how specialised a cell becomes, it starts with this shared foundation. This is why all three structures from the Kisumu hospital smear and the shamba leaf can rightly be called cells.

function, not just its name. **(c) Why is it accurate to call a red blood cell, a white blood cell, and a sukuma wiki leaf cell all 'cells', even though they look so different under the microscope?**	

Section 2: Plant vs. Animal Cells — Structural Differences and Their Reasons

The sukuma wiki leaf cell and the human cells in the blood smear are both eukaryotic cells, yet they look very different. Answer the following: **(a) Draw and LABEL a generalised plant cell and a generalised animal cell side by side. Your diagrams must each include at least SIX labelled structures.** **(b) Identify THREE structures found in the sukuma wiki leaf cell that are NOT found in the human blood cells, and explain the function of each.** **(c) For each of the three structures you identified in (b), explain WHY an animal cell does NOT need that structure. Connect your reasoning to the different lifestyles and needs of plants and animals.**	A generalised plant cell contains the following structures: cell wall, cell membrane, nucleus, cytoplasm, chloroplasts, large central vacuole, mitochondria, and ribosomes. A generalised animal cell contains: cell membrane, nucleus, cytoplasm, mitochondria, ribosomes, lysosomes, and centrioles — but lacks a cell wall, chloroplasts, and a large central vacuole. Three structures present in the sukuma wiki leaf cell but absent in human cells: 1. Cell wall — The cell wall is a rigid layer made of cellulose that surrounds the cell membrane of plant cells. It provides structural support and protection, and helps the cell maintain its rectangular shape. It also prevents the cell from bursting when it absorbs too much water. Animal cells do not need a cell wall because they have a flexible skeleton (the cytoskeleton and, for the whole body, bones and muscles) to provide support. Animals also move and need flexible membranes to allow for changes in cell shape during movement and communication. 2. Chloroplasts — Chloroplasts are green organelles that contain the pigment chlorophyll. They are the sites of photosynthesis, where light energy from the sun is used to convert carbon dioxide and water into glucose and oxygen. The sukuma wiki plant uses this glucose as its primary energy source. Animal cells do not carry out photosynthesis — animals obtain energy by consuming and digesting food (like the ugali and sukuma wiki on a student's plate), so they have no need for chloroplasts. 3. Large central vacuole — The large central vacuole in a plant cell is filled with cell sap (water and dissolved substances). It stores water, maintains turgor pressure to keep the plant firm and upright, and stores waste products. When you notice the leaf cell contents pushed to one side under the microscope, that is the vacuole taking up most of the cell's space. Animal cells may have small, temporary vacuoles, but they do not have one large permanent vacuole because animals regulate water balance through their kidneys and circulatory system rather than relying on internal water storage in each cell.

Section 3: Cell Specialisation — How Structure Matches Function

The most astonishing observation in the Kisumu hospital phenomenon was that the red blood cell had NO nucleus — a feature that seems to break the rules — while the	(a) Cell specialisation is the process by which cells develop distinct structures, shapes, and organelle compositions that allow them to carry out specific functions within an organism. Specialised cells deviate from the 'generic' cell model because certain features are enhanced, reduced, or even lost entirely to make the cell maximally efficient at its one job. (b) Three specialised cells:

white blood cell right next to it had a large, multi-lobed nucleus. Both are found in the same blood sample. This is cell specialisation in action. **(a) Define cell specialisation in your own words.** **(b) Choose THREE of the following specialised cells and, for each one, describe: (i) at least TWO structural adaptations, and (ii) how EACH adaptation directly enables the cell to perform its specific function. Choose from: red blood cell, white blood cell (neutrophil), sukuma wiki palisade mesophyll cell, root hair cell, sperm cell, nerve cell (neuron).** **(c) All specialised cells develop from a single fertilised egg cell (zygote) that contains the same DNA. If all cells have the same DNA, how is it possible that they end up looking and functioning so differently? Explain the concept of cell differentiation.**	Red Blood Cell (Erythrocyte) — Adaptation 1: Biconcave disc shape. The flattened, indented shape increases the cell's surface area relative to its volume, allowing more oxygen to diffuse in and out of the cell quickly. — Adaptation 2: No nucleus (anucleate). By losing its nucleus during maturation, the red blood cell creates more internal space for haemoglobin — the protein that actually carries oxygen. A cell packed with haemoglobin can carry far more oxygen per trip through the bloodstream. — Adaptation 3: Contains large amounts of haemoglobin. Haemoglobin binds to oxygen in the lungs and releases it in respiring tissues throughout the body, such as the muscles of a runner in a Nairobi marathon. Palisade Mesophyll Cell (Sukuma Wiki Leaf) — Adaptation 1: Tall, columnar shape packed tightly near the upper surface of the leaf. This positioning maximises the interception of sunlight for photosynthesis. — Adaptation 2: Contains a very high number of chloroplasts (up to 50 per cell). The large number of chloroplasts means the cell can absorb as much light energy as possible and produce the maximum amount of glucose for the plant. Root Hair Cell — Adaptation 1: Long, thin hair-like extension that grows out from the root. This dramatically increases the surface area in contact with soil water, allowing more water and dissolved mineral ions (like nitrates needed for protein synthesis) to be absorbed. — Adaptation 2: No chloroplasts. Root hair cells are underground and receive no light, so chloroplasts would be useless. The cell saves energy and resources by not producing them. (c) Cell differentiation is the process by which cells that all contain identical DNA become structurally and functionally different from one another. This happens because, although every cell has the full set of instructions (the entire genome), not every gene is switched ON in every cell. Specific genes are activated or silenced depending on the cell's position in the organism, the chemical signals it receives during development, and the stage of growth. For example, the gene for haemoglobin is switched ON in developing red blood cells but switched OFF in leaf cells. The gene for chlorophyll production is switched ON in plant cells exposed to light but switched OFF in root cells. The result is that each cell type produces a unique set of proteins, which in turn determines the cell's shape, the organelles it develops, and the job it can do. This is why a single fertilised egg cell can give rise to the hundreds of different cell types found in a complex organism like a human being or a sukuma wiki plant.

Section 4: Returning to the Phenomenon — Answering the Driving Question Directly	
You have now built up all the knowledge you need. Return directly to the anchoring phenomenon — the Form 4 student from Kisumu, the blood smear, and the sukuma wiki leaf slide — and answer the driving question in full:	When the Form 4 student from Kisumu looked through the microscope at their own blood smear, they saw something that seemed contradictory: red blood cells with no nucleus sitting right next to white blood cells with large, prominent nuclei. Then, when a sukuma wiki leaf slide was prepared, the cells looked completely different again — rectangular, green, and with a huge water-filled sac pushing the contents to the edge. The question is: how can all of these be called cells, and why do they look so different? All cells share a common foundation. Every cell — whether from the blood smear or the leaf — has a cell membrane to control what enters and leaves, cytoplasm where metabolic reactions occur, and genetic material (DNA) that carries the instructions for life. This shared foundation is what justifies calling all of them 'cells.' However, the reason they look so different lies in a process called cell specialisation and differentiation. Although every cell in the student's body and every cell in the sukuma wiki plant

'Why do the cells in my body — and in the sukuma wiki on my plate — look so different from each other, and how does each cell's structure allow it to do its specific job?' **Your response must:** **(a) Reference specific observations from the Kisumu blood smear and the sukuma wiki leaf slide.** **(b) Explain what cells share (the generic cell model) AND why specialised cells deviate from it.** **(c) Connect at least TWO specific structural adaptations to their functions using cause-and-effect language (e.g., 'Because… this allows… which means…').** **(d) Be written as a coherent paragraph or series of paragraphs — not a bullet list.**	contains the same DNA, different genes are switched on or off in each cell type. This causes each cell to develop unique structures — organelles and shapes — that are perfectly matched to its specific job. Because the red blood cell's main job is to transport oxygen around the body, it has been shaped by specialisation into a biconcave disc with no nucleus. The loss of the nucleus frees up space for haemoglobin, which means the cell can carry far more oxygen per journey through the bloodstream. Its flattened shape increases surface area, which allows oxygen to diffuse in and out more rapidly. Because the white blood cell's job is to defend the body against pathogens, it retains a large nucleus packed with DNA — this allows it to rapidly produce the proteins (antibodies and enzymes) needed to fight infection and to divide quickly when the body is under attack. The sukuma wiki leaf cell looks different from both blood cells because plant cells have additional structures that reflect the plant's lifestyle. Because the leaf cell must capture sunlight and carry out photosynthesis, it is packed with chloroplasts containing chlorophyll — this allows the plant to convert sunlight, water, and carbon dioxide into the glucose it needs for energy and growth. Because plants cannot move to drink water, each leaf cell maintains a large central vacuole filled with water, which keeps the cell turgid and the leaf firm and upright. The rigid cellulose cell wall gives the plant structural support in the absence of a skeleton. In summary, cells look and behave so differently because specialisation is the solution to a biological problem: complex organisms need different tasks performed simultaneously in different places. Structure always follows function. The student's blood cells and the sukuma wiki leaf cells are all built on the same cellular blueprint — but each has been elegantly modified, through millions of years of evolution and through the process of differentiation, to do its specific job as efficiently as possible.

Section 5: Reflection and Model Revision	
Look back at the cell model you drew at the very beginning of this unit (your initial model from the Predict Phase) and compare it to the final model you built across the lessons. **(a) Describe TWO specific ways your model changed over the course of the unit. What new evidence or investigation caused each change?** **(b) Identify ONE question about cells that this unit raised for you but did NOT fully answer. Explain why this question is scientifically interesting.**	(a) Two ways my model changed: First, at the start of the unit I drew a cell as a round blob with a nucleus in the middle and nothing else inside. After our microscope investigation of the sukuma wiki leaf and the onion cells, and after studying the differences between plant and animal cells, I added specific organelles to my model: chloroplasts, a large vacuole, and a cell wall for plant cells, and mitochondria, lysosomes, and ribosomes for animal cells. The evidence that caused this change was seeing the green chloroplasts clearly visible in the leaf cells under the microscope and learning that each organelle has a specific function. Second, I initially believed that all cells must have a nucleus because the nucleus contains DNA and DNA is essential for life. The Kisumu blood smear phenomenon challenged this assumption — I could see clearly that mature red blood cells had no nucleus. After the lesson on cell specialisation, I revised my model to show that the red blood cell loses its nucleus during maturation as an adaptation to maximise haemoglobin content. This taught me that the 'rules' of cell biology are shaped by function. (b) One question this unit raised but did not fully answer: How exactly does a cell 'know' which genes to switch on and which to switch off during differentiation? We learned that all cells have the same DNA but different genes are activated in different cells — but the unit did not fully explain the molecular mechanism that controls this switching. This is scientifically interesting because

(c) Scientists like Nobel Prize-winning Kenyan-linked researchers at the African Academy of Sciences continue to study cell biology to understand diseases like sickle cell anaemia, which is prevalent in communities around Lake Victoria. Sickle cell anaemia causes red blood cells to become rigid and crescent-shaped instead of the normal biconcave disc. Using what you now know about cell structure and specialisation, explain why this change in shape would prevent the red blood cell from doing its job effectively.	understanding gene regulation is at the heart of understanding how cancers develop (when cells start expressing the wrong genes and dividing uncontrollably) and how stem cell therapies might one day be used to regenerate damaged tissues. (c) In a healthy red blood cell, the biconcave disc shape is a critical adaptation: it maximises surface area for rapid oxygen diffusion and allows the cell to be flexible enough to squeeze through tiny capillaries throughout the body. In sickle cell anaemia, the red blood cells become rigid and crescent-shaped (sickle-shaped) due to an abnormal form of haemoglobin that clumps together when oxygen levels are low. This change in shape has two devastating consequences directly linked to cell structure and function. First, the reduced surface area and abnormal shape means less oxygen can diffuse into and out of the cell efficiently, so body tissues — including muscles and the brain — receive less oxygen. Second, the rigid, crescent shape means the cells cannot flex to pass through narrow capillaries. Instead, they block blood vessels, preventing any blood flow to the tissues beyond the blockage. This causes the painful 'sickle cell crises' experienced by patients. This example powerfully illustrates the core lesson of this entire unit: a cell's shape IS its function. When the structure changes, the function is lost.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of Cell Structure (Generic Cell Model)	Accurately identifies and explains all three universal cell features (membrane, cytoplasm, DNA/genetic material) with precise functional descriptions. Clearly and correctly distinguishes between plant and animal cell structures, including at least six labelled structures per diagram. Diagrams are neat, accurate, and fully labelled.	Identifies and explains at least two of the three universal cell features with mostly accurate functional descriptions. Distinguishes between plant and animal cells with at least four labelled structures per diagram. Minor errors in labelling or function descriptions are present but do not show fundamental misunderstanding.	Identifies fewer than two universal cell features, or descriptions of function are missing or inaccurate. Diagrams are incomplete, missing key structures, or labels are largely incorrect. Shows confusion between plant and animal cell structures.
Explanation of Cell Specialisation and Differentiation	Provides a clear, accurate definition of cell specialisation and differentiation. For each of three chosen specialised cells, identifies at least two structural adaptations and provides a precise, cause-and-effect explanation linking each adaptation to its function. Correctly explains that all cells share the same DNA but gene expression differs between cell types.	Provides a mostly accurate definition of cell specialisation. For at least two of three chosen cells, identifies structural adaptations and links them to function, though the cause-and-effect reasoning may not always be explicit. Understands the basic concept of differentiation but the explanation of gene switching may be incomplete.	Definition of specialisation is vague or missing. Structural adaptations are listed without being connected to function, or the connections are inaccurate. Little or no understanding of differentiation is demonstrated; the student may state that different cells have different DNA.
Direct Use of the Anchoring Phenomenon as Evidence	The response to the driving question directly and explicitly references specific	The response references the anchoring phenomenon in at least one section and	The phenomenon is mentioned by name only, or specific observations from the

	observations from both the Kisumu blood smear (e.g., biconcave shape of RBC, absence of nucleus, multi-lobed nucleus of WBC) AND the sukuma wiki leaf slide (e.g., rectangular shape, chloroplasts, large vacuole). The phenomenon is used as evidence throughout the explanation, not just mentioned in passing.	uses at least one specific observation as evidence. The connection between the phenomenon and the scientific explanation is present but may not be woven consistently throughout the response.	blood smear or leaf slide are not used as evidence. The response reads as a general biology answer disconnected from the driving question scenario.
Quality of Scientific Reasoning and Cause-and-Effect Language	Consistently uses precise cause-and-effect language (e.g., 'Because the red blood cell lacks a nucleus, it has more space for haemoglobin, which means it can transport more oxygen per cell'). Arguments are logically structured, evidence-based, and free from circular reasoning. The response tells a coherent explanatory story from evidence to conclusion.	Uses cause-and-effect language in most explanations, though some connections may be implied rather than stated explicitly. Reasoning is mostly logical and evidence-based, with minor gaps in the chain of argument. The response is generally coherent.	Cause-and-effect language is absent or rarely used. Statements are mostly descriptive ('the red blood cell has no nucleus') without explaining why this matters for function. Reasoning is incomplete, circular, or based on misconceptions.
Reflection, Model Revision, and Application to Novel Context	Clearly identifies two specific ways the cell model changed and names the evidence or investigation that drove each change, demonstrating metacognitive awareness of learning. Poses a scientifically valid and genuinely interesting unanswered question. Applies knowledge of cell structure and specialisation accurately and insightfully to the novel context of sickle cell anaemia, correctly linking the change in shape to loss of both oxygen-diffusion efficiency and capillary flexibility.	Identifies at least one specific change to the cell model with a plausible link to evidence. Poses an unanswered question that is relevant to cell biology, though it may not be deeply elaborated. Applies knowledge to sickle cell anaemia with mostly accurate reasoning, identifying at least one consequence of the shape change on cell function.	Model revision is vague or generic (e.g., 'I learned more about cells') without reference to specific evidence. Unanswered question is trivial or unrelated to cell biology. Application to sickle cell anaemia is missing, superficial, or contains significant inaccuracies about how shape affects function.